# DAY-02

# DOCUMENTATION

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## Objective:

To understand the concept of embeddings, their role in language models, and explore relevant data sources for the selected SLM project domain.

## Embeddings:

1. What are Embeddings?

   Embeddings are numerical vector representations of words, sentences, or documents that help AI models understand their meaning and relationships. They convert text into numbers so that language models can process, compare, and find similarities between different pieces of information.

2. Why are Embeddings Needed in Language Models?

- Help models understand the meaning of text.
- Capture relationships between words and sentences.
- Enable semantic search and information retrieval.
- Improve the accuracy of AI responses.
- Allow models to find similar content even when different words are used.

3. How do embeddings differ from tokenizers?

   Tokenizer:

- Splits text into smaller units (tokens).

- Prepares text for processing.
- Example: "AI is powerful"  ["AI", "is", "powerful"]
- Does not understand meaning.

Embedding:

- Converts tokens into numerical vectors.
- Represents the meaning of the text.
- Example: [0.25, 0.87, 0.44, ...]
- Captures semantic meaning and relationships.

### 4. Real-World Applications of Embeddings

- Chatbots: Understanding user queries and providing relevant responses.
- Search Engines: Finding relevant results based on meaning, not just keywords.
- Recommendation Systems: Suggesting products, movies, or content.
- Document Retrieval: Finding relevant documents from large collections.
- Question Answering Systems: Retrieving information to answer user questions.
- RAG Applications: Retrieving relevant context before generating responses.

## Selected Domain: Education

### Objective:

build a Small Language Model that helps students understand concepts, answer academic questions, and summarize study materials.

### Data Sources:

- Educational PDFs
- Study Notes

- Textbooks
- Question Banks
- Educational Articles

## Why Useful:

These resources contain academic knowledge that helps the SLM provide accurate educational assistance to students.

## Conclusion:

Embeddings help language models understand the meaning of text by converting words and sentences into numerical vectors. They play an important role in tasks such as search, retrieval, and question answering. Understanding embeddings and collecting relevant data sources are essential steps in building an effective SLM for a chosen domain.